🚀 Project Completed: Multi-Environment Infrastructure Design using Terraform Workspaces

Project Overview

In this project, I implemented a multi-environment AWS infrastructure using a single Terraform codebase. The goal was to separate Development, Staging, and Production environments while ensuring infrastructure consistency, scalability, and code reusability.

Key Steps

• Created Terraform Workspaces for Dev, Staging, and Production

• Provisioned AWS EC2 instances for each environment

• Configured Security Groups with environment-specific rules

• Created and managed AWS S3 bucket resources

• Used variables and reusable Terraform modules

• Implemented remote state management using S3 backend

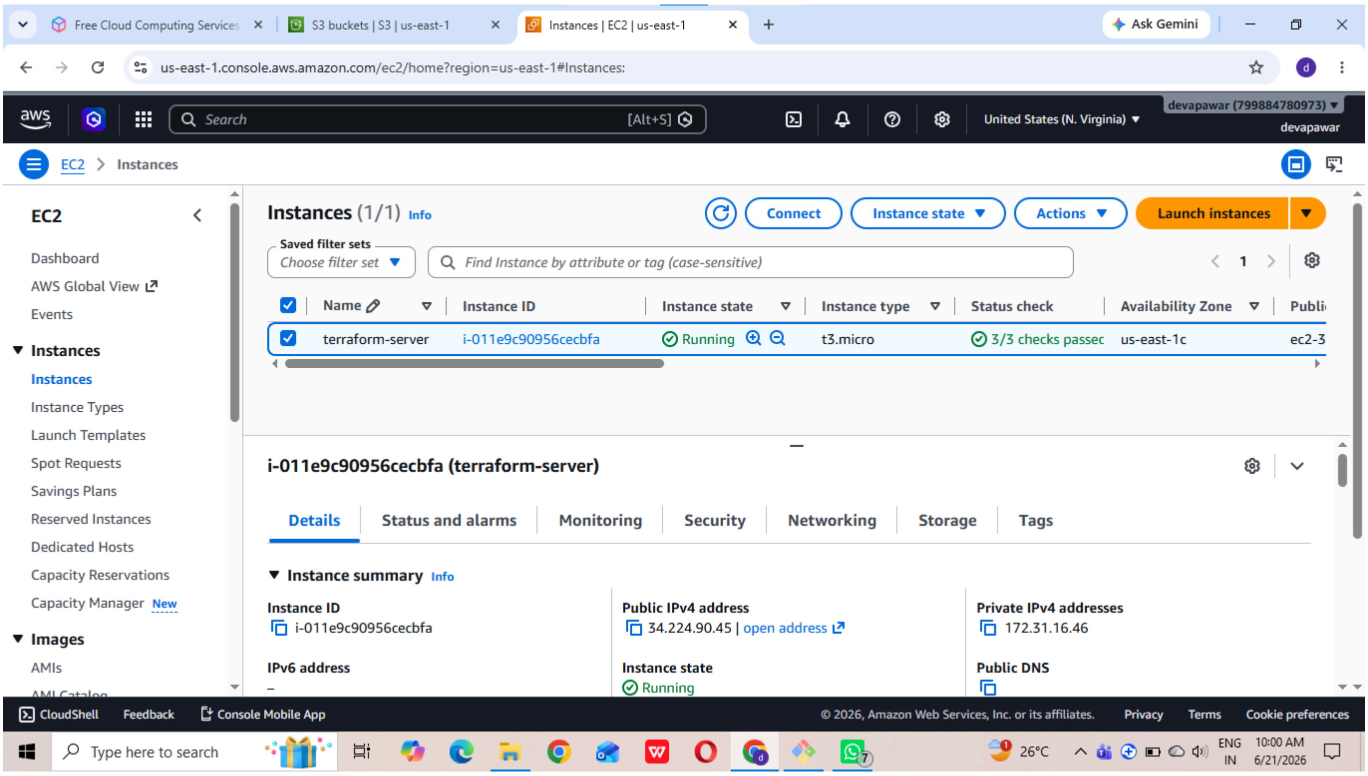
• Validated environment isolation and resource separation

Technologies Used

Terraform | AWS EC2 | AWS S3 | VPC | Security Groups | Terraform Workspaces | Infrastructure as Code (IaC)

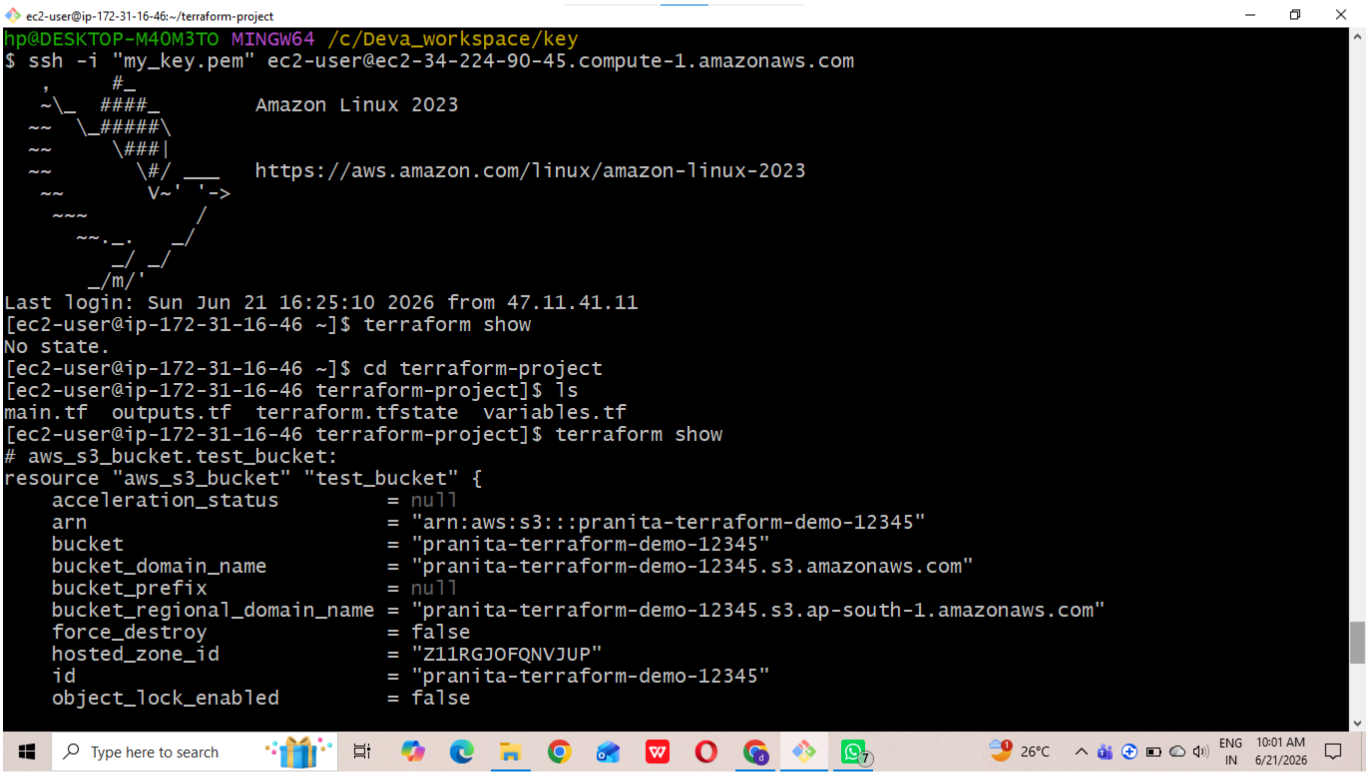
step-by-step setup Guide

STEP 1: first of all i had create ec2 instance



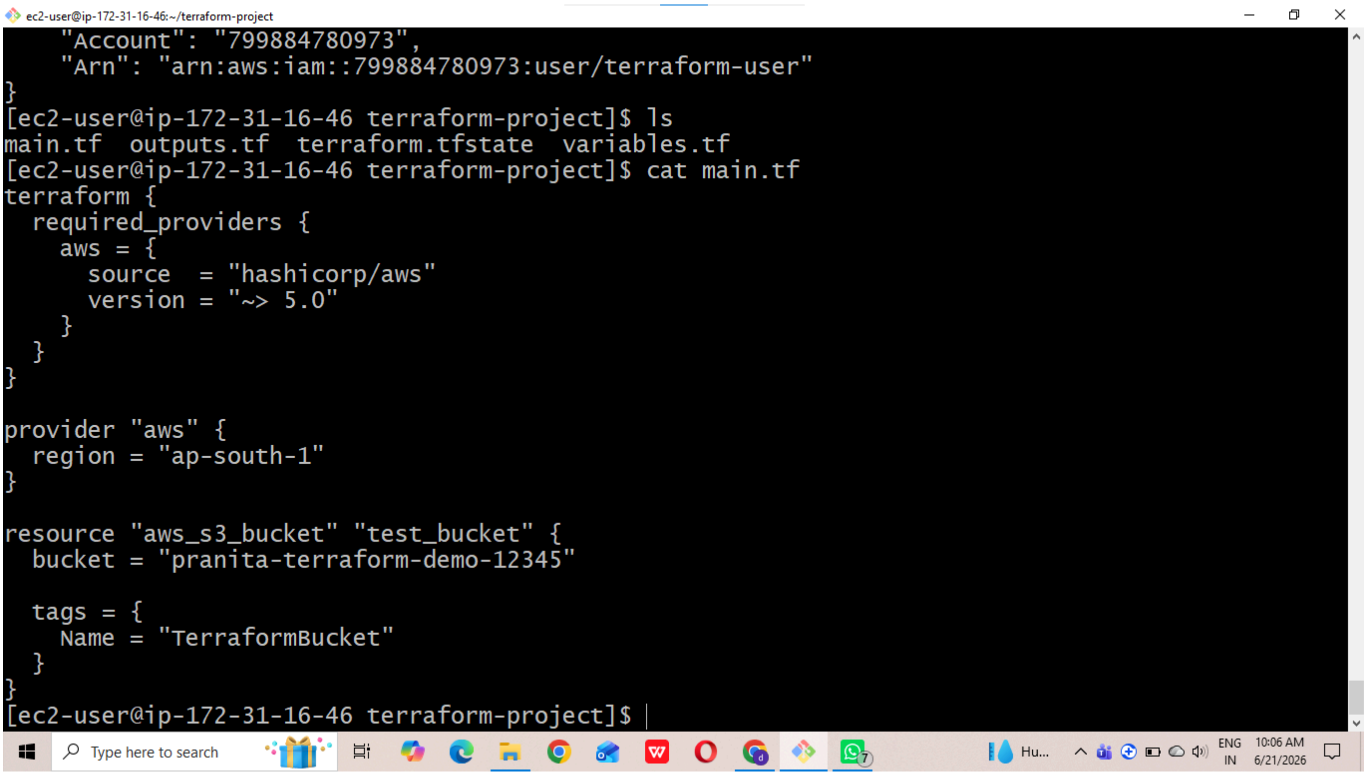
STEP2: Create a security group

STEP3: terraform workspace

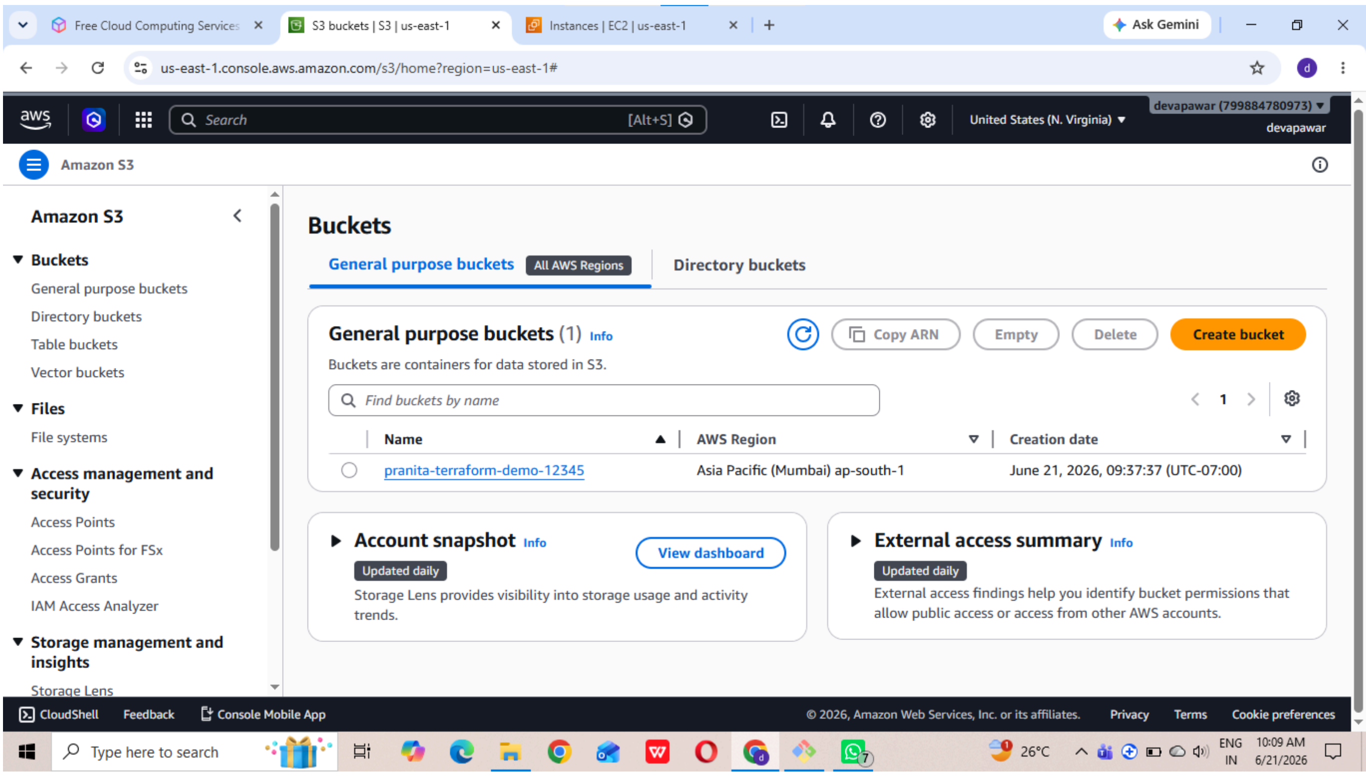


```
hp@DESKTOP-M40M3TO MINGW64 /c/Deva_workspace/key
$ ssh -i "my_key.pem" ec2-user@ec2-34-224-90-45.compute-1.amazonaws.com
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sun Jun 21 16:25:10 2026 from 47.11.41.11
[ec2-user@ip-172-31-16-46 ~]$ terraform show
No state.
[ec2-user@ip-172-31-16-46 ~]$ cd terraform-project
[ec2-user@ip-172-31-16-46 terraform-project]$ ls
main.tf  outputs.tf  terraform.tfstate  variables.tf
[ec2-user@ip-172-31-16-46 terraform-project]$ terraform show
# aws_s3_bucket.test_bucket:
resource "aws_s3_bucket" "test_bucket" {
    acceleration_status         = null
    arn                         = "arn:aws:s3:::pranita-terraform-demo-12345"
    bucket                      = "pranita-terraform-demo-12345"
    bucket_domain_name          = "pranita-terraform-demo-12345.s3.amazonaws.com"
    bucket_prefix               = null
    bucket_regional_domain_name = "pranita-terraform-demo-12345.s3.ap-south-1.amazonaws.com"
    force_destroy               = false
    hosted_zone_id              = "Z11RGJOFQNVJUP"
    id                          = "pranita-terraform-demo-12345"
    object_lock_enabled         = false
```

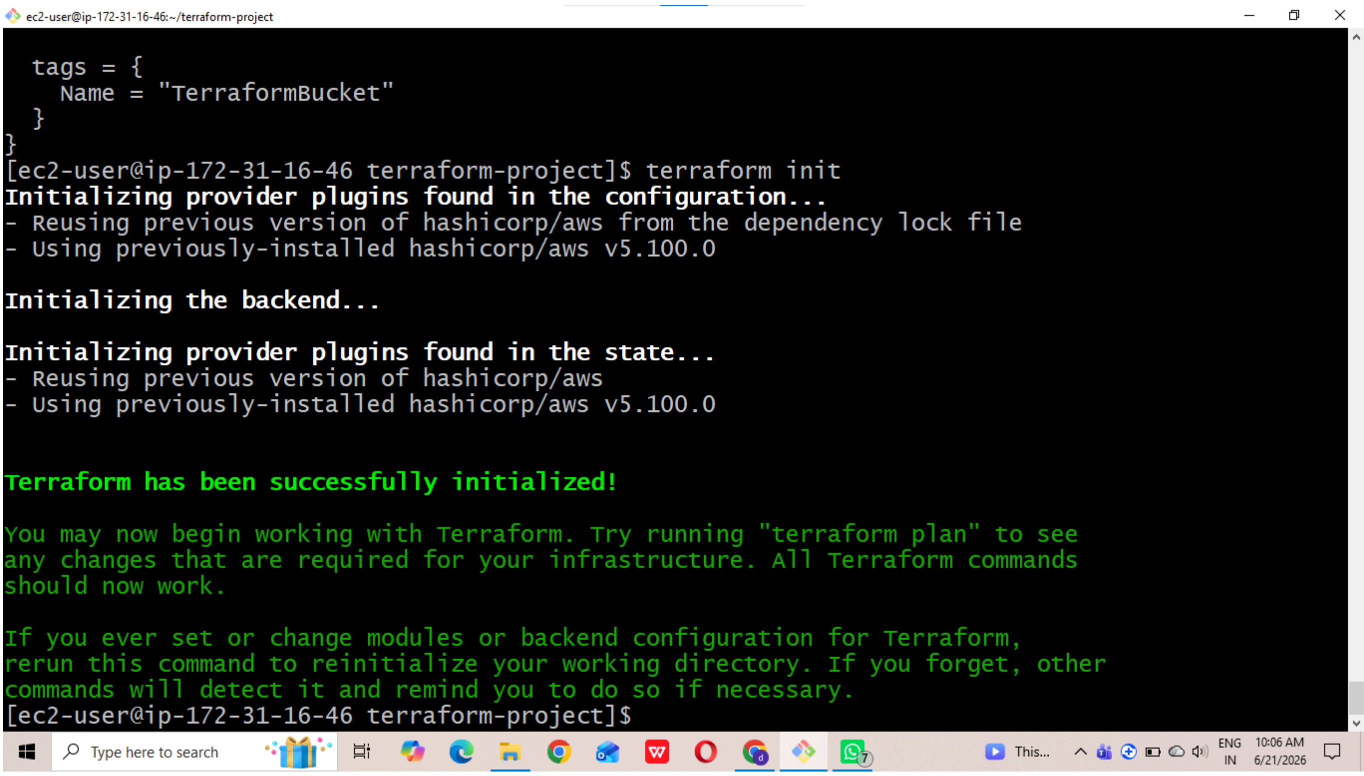
STEP4: main.tf file



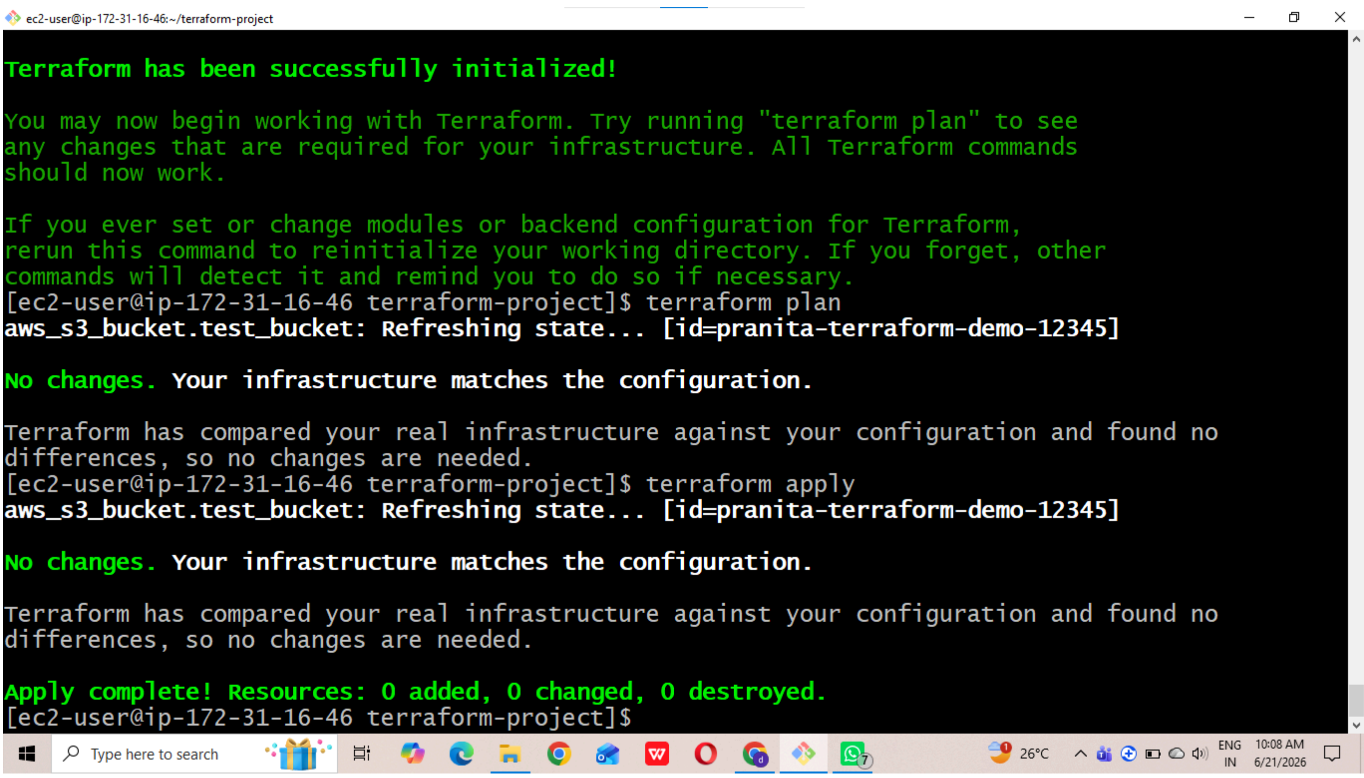
Step5: aws s3 bucket



STEP6: terraform init



STEP7:terraform apply



Project Outcome

This project helped me understand Infrastructure as Code, Terraform Workspaces, environment isolation, reusable modules, and AWS resource management. It also provided hands-on experience with managing multiple environments from a single codebase.

📌 Key Learning:

Managing Dev, Staging, and Production environments efficiently using Terraform while maintaining security, scalability, and code reusability.

#Terraform #AWS #DevOps #CloudComputing #InfrastructureAsCode #TerraformWorkspaces #EC2 #S3